

# Uganda Malaria Survey Pipeline and Risk Index

Handover report, version 2: raw DHS/MIS and GPS archives through the risk index and its validation against the malaria rapid test

Malaria Risk Index (MRI) and GeoAI project, Uganda component

Prepared for Bonny, Department of Geography and GIS, University of Cincinnati

Report date: 28 September 2026

This report replaces version 1. It records the full pipeline from the raw zipped survey and GPS archives to a scored, validated child level table. It adds the risk index, the outcome and validation work, the spatial notes on DHS coordinates, the inspection of the monthly environmental covariate sample, every defect found since version 1, and three corrections to figures and statements given earlier. Each result states how it was measured. Items that were not verified are listed in Section 14.

# 1. Summary of the current state

| Item                                                  | Value                                     | Basis                                                                          |
|-------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------|
| Children in the model input table                     | 30,486                                    | PersonRecode rows with age 0 to 5 (30,486 of 144,227 rows), four survey rounds |
| Columns in the model input table                      | 224                                       | After final selection and renaming                                             |
| Children linked to a mother                           | 20,762 (68.1%)                            | 9,724 children have no matching mother record                                  |
| Children with a valid RDT result (0 or 1)             | 23,677 (77.7%)                            | Codes 3, 4, 6, 9 excluded                                                      |
| Children scored by the risk index                     | 27,598 (90.5%)                            | Complete on all 11 inputs; 2,888 not scored                                    |
| Variance explained by the first component             | 31.6%                                     | PCA on the 27,598 complete children                                            |
| RDT positivity among scored children with a valid RDT | 29.8% (6,720 of 22,556)                   | Unweighted child counts                                                        |
| AUC of risk score against RDT result                  | 0.691 pooled; 0.641 to 0.689 within round | Within the sample used to build the score; no spatial split                    |

Open items that need a decision or more information are listed in Section 13. The three that block the next stage are the interview date source for the monthly covariate join, the scale factor of the NDVI column, and the file layout for Zimbabwe and Zambia.

# 2. Pipeline overview

Six stages run in sequence. Each writes its own CSV file, so any stage can be re run without repeating earlier ones. Stages 2 and 3 read the raw archives directly, not the stage 1 files, because stage 1 keeps only columns common to every survey round and several columns needed later exist in one round only.

| Stage | Script and function                        | Input                                | Output                                                         |
|-------|--------------------------------------------|--------------------------------------|----------------------------------------------------------------|
| 1     | dhSurvey.py<br>convert_dhs_zips_to_csv     | Raw DHS zip archives                 | Six CSV files, one per recode type                             |
| 2     | surv_index.py<br>build_malaria_index_table | Raw DHS zip archives                 | Child level CSV, 30,486 rows, 240 columns                      |
| 3     | surv_index.py<br>merge_with_gps_clusters   | Stage 2 CSV and raw GPS zip archives | Same rows with GPS covariates and coordinates, 405 columns     |
| 4     | surv_index.py<br>select_model_columns      | Stage 3 CSV                          | Model input CSV, 30,486 rows, 224 columns                      |
| 5     | build_risk_index                           | Model input CSV                      | Risk index CSV, 27,598 rows, 9 columns                         |
| 6     | merge_risk_with_outcome                    | Risk index CSV and model input CSV   | Validation CSV, 27,598 rows, risk columns plus outcome columns |

Location of stages 5 and 6. In the working copies used for testing they live in risk\_index.py. In the project repository, the traceback from the last reported run shows merge\_risk\_with\_outcome inside src/dream/surv\_index.py, so the two files have been merged there.

## 3. Data sources

### 3.1 Survey and GPS archives

| Archive              | Rounds                          | Contents relevant here                                                                                                                           |
|----------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| MIS_DHS_datasets.zip | 2009, 2014-15, 2018-19, 2024-25 | Recode files IR, HR, BR, PR, KR for all four rounds. GC (geospatial covariates) present for 2014-15, 2018-19, 2024-25 only. No GC file for 2009. |
| MIS_GPS_datasets.zip | 2009, 2014-15, 2018-19, 2024-25 | Also carries the full recode set. GC for all four rounds, including 2009. GE shapefile with raw coordinates for all four rounds.                 |

GC and GE are different files. GC holds derived environmental values for each cluster (rainfall, temperature, population, modelled malaria incidence and prevalence, and similar). It has no coordinates. GE holds LATNUM and LONGNUM. Both are needed. Point the GPS merge at the GPS archive, since the DHS archive lacks the 2009 GC file.

If both archives sit in the same raw folder, every recode type is read twice. Stage 2 removes the duplicates on the natural key of each recode type within a round, before any join, and logs the count.

### 3.2 Monthly environmental covariate sample

A sample of the separately processed monthly covariates was inspected before any join design. The file is tabular, one row per grid point and month. It is not a raster.

| Property       | Measured value                                                                                                      |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| File           | ecological_predictors_with_mri.csv, 1,048,575 data rows, 8 columns                                                  |
| Columns        | year, longitude, latitude, ndvi, month, elevation_m, precipitation_mm, temperature_C                                |
| Grid points    | 8,056 unique latitude and longitude pairs, about 130 rows per point                                                 |
| Extent         | Latitude -0.54 to 3.42, longitude 29.69 to 34.90. This is the Uganda bounding box. There is no country column.      |
| Period         | Years 2010 to 2020. The last row is September 2020. The owner confirmed the file is a sample.                       |
| Month encoding | Lower case text: jan, feb, mar, apr, may, jun, jul, aug, sept, oct, nov, dec. September is written sept.            |
| NDVI           | Integer, range 0 to 5000. The scale factor is not known.                                                            |
| Elevation      | Integer metres, 614 to 4,252. Constant within every point (0 of 8,056 points vary), as expected for a static layer. |
| Missing values | precipitation_mm: 8,032 of 1,048,575 rows (0.77%). No other column has missing values.                              |
| File name      | Contains the text mri, but there is no risk index column in the file.                                               |

The row count equals the Excel row limit. The owner confirmed that this is a sample, so the count is not treated as evidence of truncation in the full files.

## 4. Stages 1 and 2: recode extraction and the child table

### 4.1 Stage 1: recode extraction (dhSurvey.py)

Unpacks zip files nested at any depth, reads the recode type from the DHS file name (CCRRVVFL: the two letters after the country code), and combines all survey rounds of one type into one CSV. Only columns present in every round are kept, using an inner join. An earlier outer join version was killed by the operating system for memory, because the raw files have between 1,829 and 3,740 columns each and differ by round. Unreadable files and missing recode types are logged as warnings.

| File                    | Grain                | Rounds      | Rows (DHS archive) |
|-------------------------|----------------------|-------------|--------------------|
| IndividualRecode_IR.csv | One woman            | 4           | 28,226             |
| HouseholdRecode_HR.csv  | One household        | 4           | 28,771             |
| PersonRecode_PR.csv     | One household member | 4           | 144,227            |
| BirthRecode_BR.csv      | One live birth       | 1 (2009)    | 13,863             |
| KidsRecode_KR.csv       | One child under 5    | 4           | 22,824             |
| GPSClusters_GC.csv      | One cluster          | 3 (no 2009) | 892                |

### 4.2 Stage 2: child level table (build\_malaria\_index\_table)

The malaria RDT result (hml35) is recorded only for children aged 0 to 5. A first version anchored on women produced a 0% match for hml35 and was replaced. The current anchor is PersonRecode restricted to hv105 of 5 or less. Each child is left joined to her household and to her mother.

| Source                    | Join                                                          | Contributes                                                                                                               |
|---------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| PersonRecode (anchor)     | None                                                          | Child sex, age, birth order, birth interval; RDT, microscopy, PCR results and species; eight severity symptoms; treatment |
| HouseholdRecode           | hv001, hv002, survey round                                    | Nets, indoor spraying, housing materials, wealth, size, region, altitude                                                  |
| IndividualRecode (mother) | hv001, hv002, survey round; child hc60 matched to mother v003 | Mother net use, education, wealth, region; interview month and year (v006, v007)                                          |

- hc60 holds the household roster line of the mother. DHS uses 993 and 995 for no mother listed. These are set to missing before the join. Among children with a real hc60 value the match to a mother is 100%.
- Every join includes the survey round. Cluster numbers restart each round and are not comparable across rounds.
- A requested column absent from one round is logged at info level and left out for that round. It is not an error.
- Result: 9,724 of 30,486 children (31.9%) have no matching mother.

## 5. Stages 3 and 4: GPS merge and final selection

### 5.1 GPS merge (merge\_with\_gps\_clusters)

Reads the raw GPS archive a second time and attaches two things, each matched on cluster number and survey round. GC covariates come from load\_gps\_clusters. LATNUM and LONGNUM come from load\_gps\_locations, which reads the GE shapefile with geopandas. With the GPS archive, GC has 1,062 cluster and round combinations and GE has 1,062. No child was left unmatched.

### 5.2 Final selection and renaming (select\_model\_columns)

Reduces the merged table to the chosen columns and renames each DHS code. The environmental covariates are found by position, between the last GC identifier and the first GE identifier, because a GC extract can change its covariate set between downloads. Mother derived columns carry the prefix mother\_. Household columns keep plain names with an \_hr suffix where a mother copy exists. Appendix A lists the full mapping.

| Group                    | Examples                                                                                                    | Fill rate                              |
|--------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Identifiers              | cluster_number, household_number, hvidx, hc60, mother_line_number, mother_caseid                            | 100% except mother fields (68.1%)      |
| Child                    | child_sex, child_age_years, child_age_months, child_birth_order                                             | Close to 100%                          |
| Malaria outcome          | malaria_rdt_result, malaria_microscopy_result, malaria_pcr_result, species, severity symptoms, treatment    | RDT 77.7%; microscopy 77.9%; PCR 22.9% |
| Household                | household_size, wealth_score_hr, floor_material, has_mosquito_net                                           | Close to 100%                          |
| Mother                   | mother_education_level, mother_wealth_quintile, mother_net_type_last_night, interview_month, interview_year | 68.1%                                  |
| GPS cluster              | DHSID, DHSClust, DHSYEAR, LATNUM, LONGNUM                                                                   | 100% with the GPS archive              |
| Environmental covariates | Rainfall, temperature, population, modelled malaria layers, and similar benchmark year values (155 columns) | Varies by covariate and round          |

| Survey round | Children |
|--------------|----------|
| 2009         | 4,940    |
| 2014-15      | 6,108    |
| 2018-19      | 9,748    |
| 2024-25      | 9,690    |
| All rounds   | 30,486   |

## 6. Stage 5: risk index (build\_risk\_index)

The index combines household and net use variables into one continuous score between 0 and 1, then splits the score into four quartile categories. Outcome columns are never inputs, so the index can be checked against the outcome afterward without circularity.

### 6.1 Inputs

| Input                   | Source              | Coding or recode                                                                                                                                                            | Direction  |
|-------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| has_mosquito_net        | hv227               | 0 or 1                                                                                                                                                                      | Protective |
| total_nets_owned        | hml1                | Count                                                                                                                                                                       | Protective |
| net_coverage_score      | hv228               | Recoded. Raw codes: 0 none of the children under 5 slept under a net, 1 all, 2 some, 3 no net in household. Recoded to 0 no net, 1 net present but not used, 2 some, 3 all. | Protective |
| under5_net_users_count  | hml2                | Count                                                                                                                                                                       | Protective |
| floor, wall, roof tiers | hv213, hv214, hv215 | Tens digit of the DHS code: 1x natural = 3, 2x rudimentary = 2, 3x finished = 1. Code 96 (other) and any other code become missing.                                         | Risk       |
| has_electricity         | hv206               | 0 or 1                                                                                                                                                                      | Protective |
| crowding_ratio          | hv009 / hv216       | People per sleeping room. Missing when rooms is 0 or missing.                                                                                                               | Risk       |
| wealth_score_hr         | hv271               | Continuous DHS wealth factor score                                                                                                                                          | Protective |
| altitude_m_hr           | hv040               | Metres                                                                                                                                                                      | Protective |

Protective columns are multiplied by -1 before the analysis, so that a higher value means higher risk for every input.

### 6.2 Variables left out and why

| Variable                                                                 | Reason                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| malaria_*, symptom_*, first_line_medicine_*                              | Outcome columns. Keeping them out prevents circular validation.                                                                                                                                                                            |
| net1_age_months                                                          | Missing for every household without a net (0% filled) and filled for 99.9% of households with a net. A substituted value would be invented. Dropping those rows would remove the households with no net, which are the highest risk group. |
| mother_region, mother_residence_type, mother_altitude_m, mother_wealth_* | Same values as the household copies, and filled for 68.1% of children against 100%.                                                                                                                                                        |
| mother_education_level, mother_net_type_last_night                       | Filled for 68.1% of children. Including them would shrink the scored sample. Candidates for an extended index.                                                                                                                             |
| child_sex, child_age, child_birth_order, child_preceding_birth_interval  | Attributes of the child, not of the household. Reserved as covariates for models that use the score.                                                                                                                                       |
| region_hr, residence_type_hr                                             | Coarse administrative categories. The monthly environmental covariates will carry this information more precisely.                                                                                                                         |
| wealth_quintile_hr                                                       | Binned copy of wealth_score_hr.                                                                                                                                                                                                            |
| household_size, sleeping_rooms_count (separately)                        | Combined into crowding_ratio.                                                                                                                                                                                                              |
| children_under5_count, cooking_fuel_type                                 | Weak or ambiguous link to malaria risk.                                                                                                                                                                                                    |
| Identifiers, sample weights, survey_round, interview month and year      | No risk content.                                                                                                                                                                                                                           |

## 6.3 Method

- 1 Read the model input CSV. Recode the materials and net coverage and derive the crowding ratio. If these derived columns already exist, for example when the function is run on its own earlier output, replace them and log the names.
- 2 Negate the protective columns.
- 3 Keep children that have all 11 inputs. No value is imputed.
- 4 Standardize each input to mean 0 and standard deviation 1. Fit PCA with one component.
- 5 Fix the arbitrary sign of the component by its correlation with crowding\_ratio, so that a higher score always means higher risk.
- 6 Rescale the component to the range 0 to 1 using its minimum and maximum (raw range -5.658 to 4.903).
- 7 Cut the scored children into quartiles labelled extremely low risk, low risk, high risk, extremely high risk.
- 8 Drop children without a score and write nine columns.

## 6.4 Result

27,598 of 30,486 children have all inputs (90.5%). The 2,888 others are missing at least one input. The largest gaps measured in the inputs are the hv228 based net coverage (5.4% of children missing) and wealth\_score\_hr (2.1% missing). The first component explains 31.6% of the variance.

| Input                  | Loading on component 1 |
|------------------------|------------------------|
| wealth_score_hr        | +0.446                 |
| floor_material_tier    | +0.396                 |
| wall_material_tier     | +0.377                 |
| roof_material_tier     | +0.370                 |
| has_electricity        | +0.329                 |
| total_nets_owned       | +0.279                 |
| net_coverage_score     | +0.251                 |
| has_mosquito_net       | +0.249                 |
| under5_net_users_count | +0.181                 |
| crowding_ratio         | +0.129                 |
| altitude_m_hr          | +0.067                 |

Loadings are on the negated, standardized inputs. All have the same sign. The wealth score and the three housing materials carry the largest weights, so the score is mostly a socioeconomic and housing quality axis. Net variables load at 0.18 to 0.28. Altitude and crowding load weakly.

| Category            | Score range    | Children scored |
|---------------------|----------------|-----------------|
| extremely low risk  | 0.000 to 0.412 | 6,901           |
| low risk            | 0.412 to 0.560 | 6,899           |
| high risk           | 0.560 to 0.660 | 6,898           |
| extremely high risk | 0.660 to 1.000 | 6,900           |

## 6.5 Output file

Nine columns: cluster\_number, household\_number, hvidx, mother\_caseid, survey\_round, LATNUM, LONGNUM, malaria\_risk\_score, malaria\_risk\_category. The first three with survey\_round identify the child. mother\_caseid is kept but does not identify a child, because siblings share it.

## 7. Outcome and validation

### 7.1 Choice of outcome

The primary validation outcome is malaria\_rdt\_result (hml35). It is present in all four rounds and it is the field DHS uses for its reported prevalence indicator. It must be restricted to 0 and 1 before use. In the scored sample 18 children carry other codes.

| Column (DHS code)                  | Values                                                                                       | Coverage                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| malaria_rdt_result (hml35)         | 0 negative, 1 positive, 3 not present, 4 refused, 6 other. Code 9 also occurs in the data.   | 23,677 valid of 30,486 (77.7%)                              |
| malaria_microscopy_result (hml32)  | 0 negative, 1 positive, 5 refused, 6 other, 7 undetermined                                   | 77.9% not missing                                           |
| malaria_measurement_status (hml33) | 0 measured, 3 not present, 4 refused, 6 other. This is a completion flag, not a test result. | Not an outcome                                              |
| malaria_pcr_result (sPCR)          | 0 negative, 1 positive; other codes                                                          | 22.9%. Exists for 2024-25 only.                             |
| symptom_* (eight columns)          | Severity symptoms                                                                            | Any of the eight present for 23.2% of RDT positive children |

hml33 was first named as a combined test result, which was wrong. The project table uses malaria\_measurement\_status. The working copy of surv\_index.py used for testing in this report still maps hml33 to the old name, so the master copy should be checked.

### 7.2 No composite outcome index

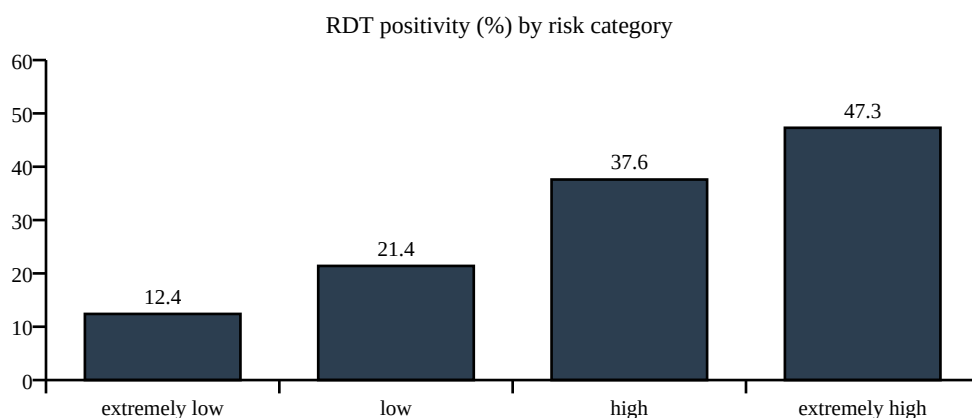
A composite outcome was considered and rejected. The severity symptoms exist for 23.2% of RDT positive children, so a severity scale would rest on under a quarter of the positives. RDT and microscopy disagree often enough that combining them by either test changes what is measured:

|              | Microscopy negative | Microscopy positive | Total  |
|--------------|---------------------|---------------------|--------|
| RDT negative | 16,209              | 449                 | 16,658 |
| RDT positive | 2,885               | 4,111               | 6,996  |
| Total        | 19,094              | 4,560               | 23,654 |

Children with a valid 0 or 1 result on both tests. 41.2% of RDT positive children are microscopy negative. 2.7% of RDT negative children are microscopy positive. RDT detects an antigen that can persist after an infection clears, so the two tests are not interchangeable. Report the concordance as a sensitivity note and keep RDT as the outcome.

### 7.3 Validation of the score against the RDT

Measured on the 22,556 scored children with a valid RDT result. Percentages are unweighted child counts and are not national prevalence estimates.



| Risk category       | Children with valid RDT | RDT positive |
|---------------------|-------------------------|--------------|
| extremely low risk  | 5,500                   | 12.4%        |
| low risk            | 5,713                   | 21.4%        |
| high risk           | 5,689                   | 37.6%        |
| extremely high risk | 5,654                   | 47.3%        |
| All                 | 22,556                  | 29.8%        |

| Survey round | Children with valid RDT | RDT positive | AUC   |
|--------------|-------------------------|--------------|-------|
| 2009         | 3,924                   | 53.2%        | 0.641 |
| 2014-15      | 4,677                   | 33.1%        | 0.648 |
| 2018-19      | 7,605                   | 21.0%        | 0.689 |
| 2024-25      | 6,350                   | 23.4%        | 0.681 |
| Pooled       | 22,556                  | 29.8%        | 0.691 |

Positivity rises at every step across the four categories. The AUC of 0.691 is moderate. Prevalence differs strongly between rounds, so the pooled gradient mixes between round and within round differences. The within round AUC values, 0.641 to 0.689, are the cleaner evidence. All figures come from the same children used to build the score, without a spatial or temporal split.

### 7.4 Sensitivity check on the wealth score

DHS computes the wealth factor score inside each survey. The score has the largest loading, so pooling it over four rounds was checked. Its mean and spread by round are close (means -10,659 to -13,713; standard deviations 89,137 to 95,805). Standardizing the score within each round before the analysis left the pooled AUC at 0.691. The 2024-25 AUC moved from 0.681 to 0.680 and the other three round values were unchanged at three decimals. This check was run outside the pipeline and is not part of it.

## 7.5 Joining the score to the outcome (merge\_risk\_with\_outcome)

The risk file has no outcome columns by design. The function joins them back from the model input table on cluster\_number, household\_number, hvidx, and survey\_round together. Do not join on mother\_caseid: 5,623 of 14,425 mother and round combinations have more than one child row, up to five, so that join is many to many. The function uses the merge indicator to count keys that fail to match, and reports separately how many matched children have no outcome value.

## 8. Spatial notes

### 8.1 Why coordinates repeat

Coordinates belong to the cluster. Rows belong to the child. There are 1,062 cluster and round combinations, with a mean of 28.7 children each (range 3 to 63), and 1,037 distinct coordinate pairs. A few physical clusters were surveyed in more than one round and share a point. Children in one cluster share environment and are not independent observations. Split training and test data by cluster, not by child.

### 8.2 What the DHS point represents

DHS does not release cluster boundaries or areas. Each point is displaced at random before release. The commonly used practice is to average covariates over a buffer equal to the displacement radius. This follows the DHS displacement procedure as summarized in sources retrieved by web search during the project. The DHS page itself was not re fetched for this report.

| Cluster type                  | Maximum displacement | Buffer area       |
|-------------------------------|----------------------|-------------------|
| Urban                         | 2 km                 | about 12.6 sq km  |
| Rural (99% of rural clusters) | 5 km                 | about 78.5 sq km  |
| Rural (1% of rural clusters)  | 10 km                | about 314.2 sq km |

In the model input, 284 cluster and round combinations are urban and 756 are rural. A third residence code appears for 22 combinations in 2018-19. Its meaning was not traced. For the monthly covariate join, take the mean of grid points inside a 2 km (urban) or 5 km (rural) buffer instead of the single nearest point.

## 9. Defects found and how they were resolved

Each item was found by running the code against the archives or by running a checker, not by reading the code.

| Defect                                             | Cause                                                                  | Resolution                                                                             |
|----------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Process killed while combining rounds              | Outer join over files with 1,829 to 3,740 columns that differ by round | Inner join in stage 1. Stage 2 selects a fixed column list per round before combining. |
| Row counts doubled, then multiplied by later joins | Both archives in one folder carry the full recode set                  | Deduplicate each recode type on its natural key and round, before joining              |
| No 2009 covariates                                 | DHS archive has no 2009 GC file                                        | Use the GPS archive                                                                    |
| No coordinates                                     | GC holds covariates, not coordinates                                   | GE shapefile read with geopandas                                                       |
| hml35 empty for every woman                        | RDT is recorded only for ages 0 to 5                                   | Anchor on children                                                                     |

| Defect                                                   | Cause                                                                                                                                       | Resolution                                                                              |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Apparent 32% non match to mothers, dtype warning         | hc60 sentinel codes 993 and 995                                                                                                             | Set to missing before the join. 100% match for real line numbers.                       |
| Backward loading for net coverage                        | hv228 was treated as ordinal. Its numeric order (0 none, 1 all, 2 some, 3 no net) is not the coverage order                                 | Explicit recode to a 0 to 3 coverage scale                                              |
| net1_age_months unusable in the index                    | Missing exactly where no net exists                                                                                                         | Left out of the index                                                                   |
| hml33 read as a test result                              | It is a measurement completion status                                                                                                       | Renamed by the owner. Not used as an outcome.                                           |
| Join of risk file to outcome returned 21,177,730 rows    | mother_caseid is shared by siblings, so the join was many to many                                                                           | The risk file carries the child key. Join on the child key. 27,598 rows in, 27,598 out. |
| Warning said keys did not match when they did            | Missing outcome was used as a sign of a failed join. 5,024 children matched but were never tested                                           | Use the merge indicator. Separate info line for missing outcomes.                       |
| Crash when the risk function read its own earlier output | Derived columns already present were duplicated                                                                                             | Detect, replace, and log the column names                                               |
| KeyError on cluster_number in the owner run              | The risk file on disk was written by an older build_risk_index (6 columns), and read by the newer merge function                            | Replace both functions together. Regenerate the risk file.                              |
| Automatic formatter made a dictionary layout worse       | autopep8 aggressive mode collapsed a long literal into staircase indentation                                                                | Output discarded. Fixed by hand with pycodestyle as the check.                          |
| PEP 8 violations                                         | 63 reported in surv_index.py (E128, E501, E203, E303, W292), 5 in risk_index.py (E501), missing final newline in dhSurvey.py and run_dhs.py | Fixed. pycodestyle at default settings reports none.                                    |
| Unreachable duplicate return statement                   | Editing slip at the end of merge_risk_with_outcome                                                                                          | Removed                                                                                 |

## 10. Corrections to figures and statements given earlier

| Earlier statement                                                              | Correction                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RDT positivity by risk category was 12.6%, 22.1%, 38.5%, 47.7%.                | Those means used the raw RDT column, which holds 18 scored children with non result codes (3, 4, 6, 9). Restricted to 0 and 1 the figures are 12.4%, 21.4%, 37.6%, 47.3%. The pattern is unchanged.                                                                                        |
| All delivered files follow the single quote convention.                        | Not true for the current files. Counts of double and single quoted strings outside docstrings: dhSurvey.py 38 and 0, surv_index.py 169 and 239, run_dhs.py 5 and 12, risk_index.py 0 and 87. PEP 8 accepts either quote if one rule is used consistently. pycodestyle does not check this. |
| Interview month and year (v006, v007) will support the monthly covariate join. | They come from the mother record and are filled for 68.1% of children (65.5% to 70.7% by round). HR and PR carry the household interview month and year (hv006, hv007) with 100% fill. This was checked on the 2024-25 files.                                                              |

## 11. Known limitations

- Mother linkage is 68.1%. Mother derived columns are missing for the rest.
- GC covariates are benchmark year values, not a monthly series. The owner's monthly NDVI, precipitation, and temperature files are not joined yet.
- Interview month and year in the table cover 68.1% of children (see Section 10).
- PCR exists for 2024-25 only. Symptoms exist for 23.2% of RDT positive children.
- The first component explains 31.6% of variance and is mostly a socioeconomic and housing axis. Nets and altitude have small weights.
- Validation is within the sample used to build the score. AUC is 0.691. No cluster level or round level hold out was run.
- 2,888 children (9.5%) are not scored. Whether they differ by round or region was not tested.
- All percentages are unweighted. DHS sample weights (household weight hv005 and mother weight v005, both divided by 1,000,000 before use) are in the model input table and were not applied.
- DHS displaces coordinates by up to 2 km (urban), 5 km (rural), or 10 km (1% of rural). Point values carry that positional error.
- The stage 2 file carries birth history fields repeated by birth slot (h22\_1 to h22\_6 and similar). They belong to the mother, not to the specific child, and were not matched to a child. They are dropped in the model input table.
- `surv_index.py` imports `geopandas` at module level without a guard, so the whole module fails to import if `geopandas` is missing. An earlier version degraded to a warning.
- Quote style is mixed across files (Section 10).

## 12. Next steps

### 12.1 Before the monthly covariate join

- 1 Add hv006 and hv007 from HR (or PR) to the child table, so that every child has an interview month and year. Without this, 31.9% of children cannot be placed in time.
- 2 Confirm the NDVI scale factor for the 0 to 5000 integer column.
- 3 Confirm the layout of the Zimbabwe and Zambia covariate files: one file per country with the same eight columns, or another layout. Add a country label, since the sample has none.
- 4 Map month names to numbers, including sept.
- 5 Decide how to treat the 0.77% missing precipitation values. Do not fill them with a default.

## 12.2 Join design

- Spatial: mean of covariate grid points inside a 2 km buffer for urban clusters and a 5 km buffer for rural clusters, using residence\_type\_hr. A nearest point join is the fallback.
- Temporal: match the interview month and year, and test lags of 1 to 3 months. Rainfall and temperature act on transmission with a delay. Report the lag chosen and the reason.
- Elevation is static. Join it once per cluster.

## 12.3 Modelling and validation

- Split by cluster or by region, not by child, to avoid leakage between siblings and neighbours.
- Use malaria\_rdt\_result restricted to 0 and 1. Report RDT against microscopy concordance as a sensitivity note.
- Report the score both pooled and within round. Apply DHS weights for any prevalence figure.
- Treat severe malaria as an exploratory analysis, using the WHO criteria that map to the symptom columns, with the 23.2% coverage stated.

## 12.4 Research plan (not started)

- Index paper: construction, external validation, and the sensitivity checks in this report.
- Spatiotemporal forecasting model using the child level table and the monthly covariates.
- Synthetic data for privacy preserving sharing: tabular synthesis for survey variables and geostatistical simulation for spatial covariates.
- Data sovereignty quantification: compare pooled training, federated training, and synthetic data sharing, and report the performance retained relative to pooled training. Zambia takes part in this thread only, as a new federation member.
- Transfer learning from Uganda to Zimbabwe, with zero shot, fine tuned, and Zimbabwe only baselines. The owner holds DHS data for all three countries.

## 13. Open items needing a decision

| Item                                | Decision or information needed                                                                                    |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Interview date source               | Approve adding hv006 and hv007 from HR or PR to the child table.                                                  |
| NDVI scale factor                   | State the divisor for the 0 to 5000 values.                                                                       |
| Zimbabwe and Zambia covariate files | State the file layout and country labelling.                                                                      |
| hml33 name                          | Confirm malaria_measurement_status in the master copy of surv_index.py.                                           |
| Extended index                      | Decide whether to add mother_education_level and mother_net_type_last_night, accepting the smaller scored sample. |
| Pooled or per round index           | Decide whether the index is fitted on all rounds together or per round.                                           |
| Quote style                         | Decide whether to convert dhSurvey.py and surv_index.py to single quotes, as the project rule states.             |
| geopandas import                    | Decide whether to restore the guarded import.                                                                     |

## 14. Verification record

### 14.1 Verified, and how

| Claim                                                           | Method                                                                                                                                        |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Row and column counts in Sections 1, 2, 5, 6                    | Full pipeline run on the two sample archives after each change                                                                                |
| Join keys are unique in the model input table                   | Grouped count on (cluster_number, household_number, hvidx, survey_round): zero duplicates                                                     |
| Risk file joins to the model input table completely             | Merge indicator: 27,598 both, 0 left only                                                                                                     |
| Value codes for hv213, hv214, hv215, hv228, hml35, hml32, hml33 | Read from the value labels in the raw Stata files (2024-25 round)                                                                             |
| Validation figures in Section 7                                 | Recomputed from the model input table with strict 0 and 1 RDT values                                                                          |
| Wealth score sensitivity (Section 7.4)                          | Separate script, standardized within round, AUC compared                                                                                      |
| Interview date coverage (Section 10)                            | Fill rates read from the model input and from the 2024-25 HR and PR files                                                                     |
| Style                                                           | pycodestyle at default settings, 79 characters, run on dhSurvey.py, surv_index.py, risk_index.py, and run_dhs.py: no reports at the last edit |

### 14.2 Not verified

- Execution on the owner's machine (Windows, src/dream layout). The one reported error was traced to a stale risk file. The fix was not run there.
- Behaviour of the pipeline without geopandas installed.
- Any spatial or temporal hold out of the risk score.
- Stability of the loadings across rounds. The PCA was fitted on the pooled rounds.
- The full monthly covariate files. Only the 1,048,575 row Uganda sample was inspected.
- The Zimbabwe and Zambia survey archives. The pipeline was run on the Uganda archives only.
- The meaning of residence code 3 in 2018-19.
- Application of DHS sample weights.

## 15. Files, conventions, and how to run

| File          | Purpose                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------|
| dhSurvey.py   | Zip extraction, recode type detection, survey round label, file loading, and the six file extractor |
| surv_index.py | Stages 2 to 4, the column lists and rename map                                                      |
| risk_index.py | Stages 5 and 6 and their helpers (merged into surv_index.py in the repository)                      |
| run_dhs.py    | Entry point. Reads script_config.ini and runs the stages in order                                   |

```

build_malaria_index_table(base_dir=DATA_RAW, output_path=OUTPUT_PATH)
merge_with_gps_clusters(OUTPUT_PATH, GPS_RAW, MERGED)
select_model_columns(MERGED, REFINED)
build_risk_index(REFINED, RISK_INDEXED)
merge_risk_with_outcome(RISK_INDEXED, REFINED, VALIDATION)

```

Dependencies: pandas, numpy, scikit-learn, geopandas. pycodestyle is used for the style check only. After replacing the risk functions, delete and regenerate the risk index CSV, since an older file lacks the child key columns.

## Conventions in force

- Module header with Role, Description, and Author. NumPy style docstrings. Docstring lines at 72 characters or less.
- PEP 8 with a 79 character line limit. Four space indents. Hanging indents for wrapped calls. No spaces around the equals sign of a keyword argument.
- The logging module in library code, with percent placeholders. No print in library code.
- Specific exception classes. No substituted values for missing data. A value that cannot be determined is left missing and logged.
- Single quotes by default, double quotes when the string contains an apostrophe, and triple double quotes for docstrings (Section 10 lists the files that do not yet follow this).

## Appendix A: column rename map

| Code   | Name                       | Code   | Name                           |
|--------|----------------------------|--------|--------------------------------|
| hv001  | cluster_number             | hv214  | wall_material                  |
| hv002  | household_number           | hv215  | roof_material                  |
| v003   | mother_line_number         | hv206  | has_electricity                |
| caseid | mother_caseid              | hv226  | cooking_fuel_type              |
| ml101  | mother_net_type_last_night | hml4_1 | net1_age_months                |
| v005   | mother_sample_weight       | hv104  | child_sex                      |
| v024   | mother_region              | hv105  | child_age_years                |
| v025   | mother_residence_type      | hml16a | child_age_months               |
| v040   | mother_altitude_m          | hc64   | child_birth_order              |
| v106   | mother_education_level     | hc63   | child_preceding_birth_interval |
| v190   | mother_wealth_quintile     | hml35  | malaria_rdt_result             |
| v191   | mother_wealth_score        | hml32  | malaria_microscopy_result      |
| v006   | interview_month            | hml32a | malaria_species_falciparum     |
| v007   | interview_year             | hml32b | malaria_species_malariae       |
| hhid   | household_id               | hml32c | malaria_species_ovale          |
| hv227  | has_mosquito_net           | hml32d | malaria_species_vivax          |
| hml1   | total_nets_owned           | hml33  | malaria_measurement_status*    |

| Code  | Name                        | Code   | Name                          |
|-------|-----------------------------|--------|-------------------------------|
| hv228 | under5_slept_net_last_night | spr    | malaria_pcr_result            |
| hml2  | under5_net_users_count      | sspec  | malaria_pcr_species           |
| hv005 | household_sample_weight     | hml37a | symptom_extreme_weakness      |
| hv024 | region_hr                   | hml37b | symptom_heart_problems        |
| hv025 | residence_type_hr           | hml37c | symptom_loss_of_consciousness |
| hv040 | altitude_m_hr               | hml37d | symptom_rapid_breathing       |
| hv270 | wealth_quintile_hr          | hml37e | symptom_seizures              |
| hv271 | wealth_score_hr             | hml37f | symptom_abnormal_bleeding     |
| hv009 | household_size              | hml37g | symptom_jaundice              |
| hv014 | children_under5_count       | hml37h | symptom_dark_urine            |
| hv216 | sleeping_rooms_count        | hml38  | first_line_medicine_given     |
| hv213 | floor_material              | hml39  | first_line_medicine_accepted  |

\* Name in the project table. The working copy used for testing maps hml33 to malaria\_test\_composite. See Section 7.1.

---

End of report. It records the pipeline as of the last working session and should be regenerated after further changes.